MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPES SR-554 AND 554

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Clipper diode or rectifier, high vacuum

See figure 1

Mounting position: Any

Weight: 1.2 ounces (34 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib	Irms	tk
Unit:	Vac	kv	a	mAdc	mAac	sec
Clipper:						
Maximum:	7.2	16	12	---	450	---
Minimum:	5.7	---	---	---	---	---
Rectifier:						
Maximum:	7.2	17	0.47	150	---	---
Minimum:	5.7	---	---	---	---	---
TEST CONDITIONS:	6.3	---	---	---	---	60

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part [illegible]								
1301	Heater current	-		0.65	II	If	3.0	4.0	Aac
1353	Operation of rectifier (1)	2	t = 120 sec (min)	0.65	II	---	---	---	---
1256	Electrode current (anode	-	Eb = 130 Vdc	0.65	II	Ic	100	---	mAdc
1231	Peak emission	-	eb = 4,000 v	0.65	II	is	12	---	a
	Quality conformance inspection, part [illegible]								
1031	High-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1041	Shock	1	300 G at 0.5 ms	---	---	---	---	---	---
---	Shock test end point								
1353	Operation of rectifiers (1)	-		---	---	---	---	---	---
1353	Operation of rectifiers (2)	3	Modulator shunt diode, t = 300 sec (min	---	---	ib	10	---	a
	Quality conformance inspection, part [illegible]								
---	Life test (1)	-	Group D, operation of rectifiers (1), t = 500 hours	---	---	---	---	---	---
---	Life-test (1) end points								
1231	Peak emission	-		---	---	is	10	---	a
1353	Operation of rectifiers (1)			---	---	---	---	---	---
---	Life test (2)	4	Group D, operation of rectifiers (2), t = 200 hours	---	---	---	---	---	---
---	Life-test (2) end points								
1231	Peak emission	-		---	---	is	10	---	a
1353	Operation of rectifiers (2)	-		---	---	ib	10	---	a

NOTES:

1. The tube shall be rigidly supported at the anode.
2. The tube shall be operated as a rectifier at the following conditions: epx = 17 kv; Ik = 150 mAdc, ik = 470 ma. The ambient temperature of the liquid coolant shall be 155° ±20°C. There shall be no evidence of arcing or gas.
3. The tube shall be operated as a shunt diode in a thyratron modulator under the following circuit conditions: tp = 1.0 ±0.15 µs; network impedance = 50 ±4 ohms, series diode resistor = 1,100 ohms, load resistor = 0 ohms, epy = 16 kv; pri[illegible] ambient temperature of the liquid coolant shall be 155° ±20°C. There shall be no evidence of arcing or gas during the last 2 minutes of the test interval.
4. On life test, the tube shall be operated alternately for 1 minute with load resistor at 10 ohms and 3 minutes with load resistor at 60 ohms.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

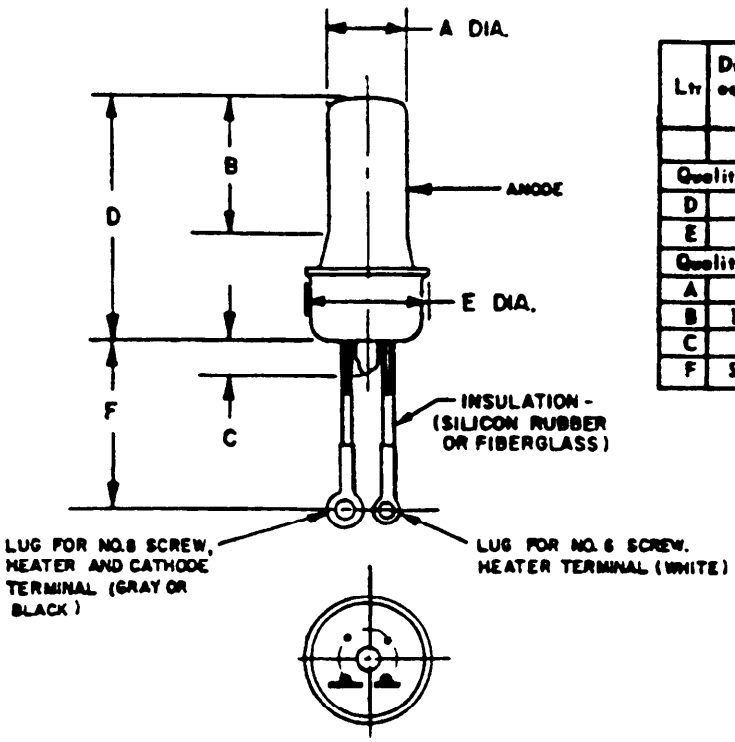
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2953)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
D		2.350 (59.69)
E		1.193 (30.30)
Quality conformance inspection, part 2		
A	.675 (17.15)	.690 (17.53)
B	1.00 (25.40)	1.20 (30.48)
C		.50 (12.70)
F	See notes a and b	

NOTES:

a. Type SR-554 lead length shall be 3±1/4 (76.2 mm ±6.35 mm) inches.

b. Type 554 lead length shall be 1.500 (38.10 mm) inches maximum.

FIGURE 1. Outline drawing of electron tube types SR-554 and 554.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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